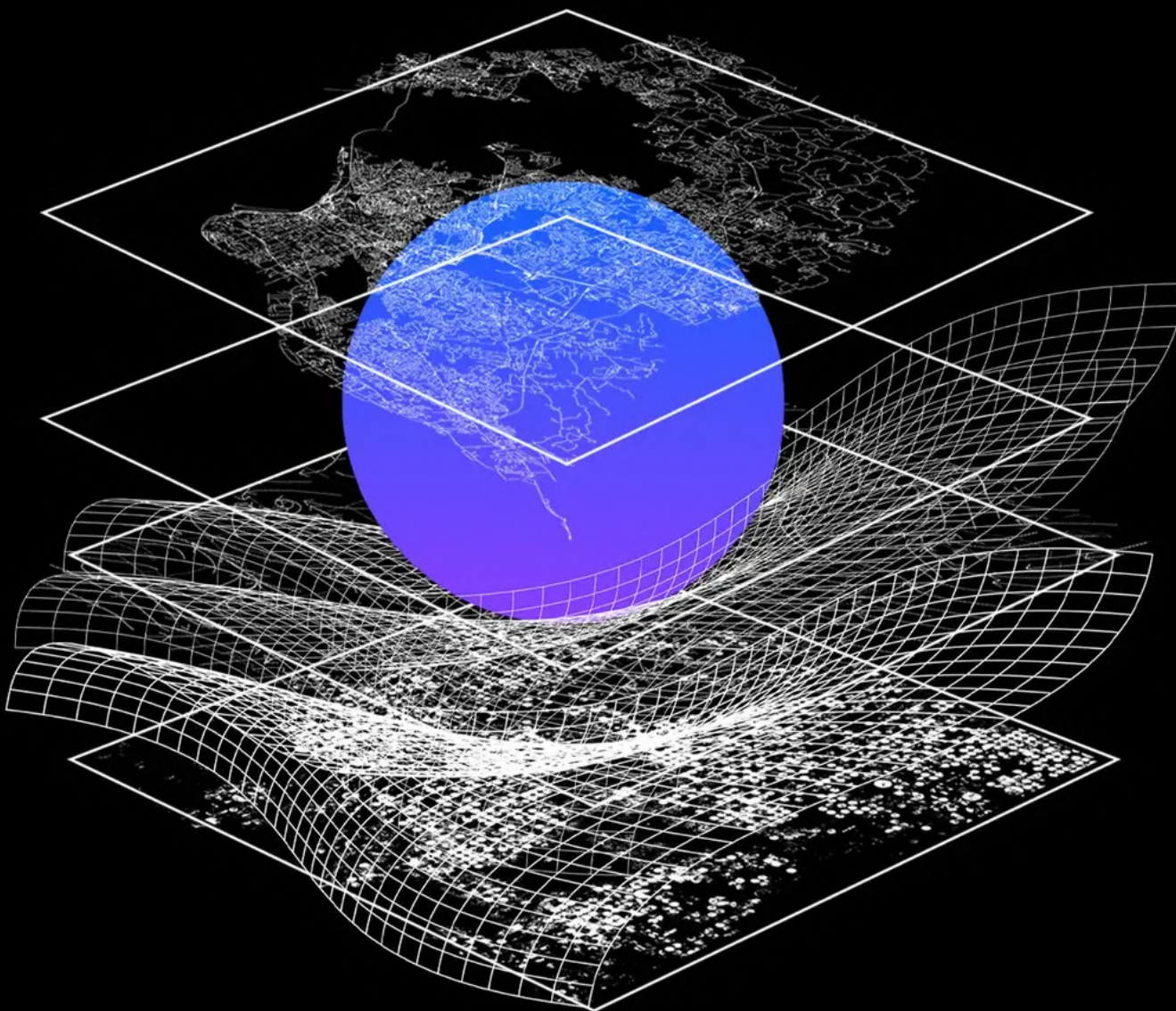




Introduction to Foundation Models

FFT

Oscar José Pellicer Valero

Carmen Oliver Huidobro

← Image taken from IBM Research Blog at:
www.research.ibm.com/blog/full-stack-foundation-models

Outline

1. Introduction to Foundation Models

- Definition, Significance, Scale, Capabilities

2. General Purpose FMs

- **Language:** GPT-3, LLMs, DeepSeek R1
- **Multimodal:** CLIP
- **Vision-Language:** LLaVA
- **Vision:** DINOv1/v3

3. Foundation Models in Earth Observation

- DINOv3 Sat
- Terramind
- AlphaEarth
- OlmoEarth

4. Conclusions

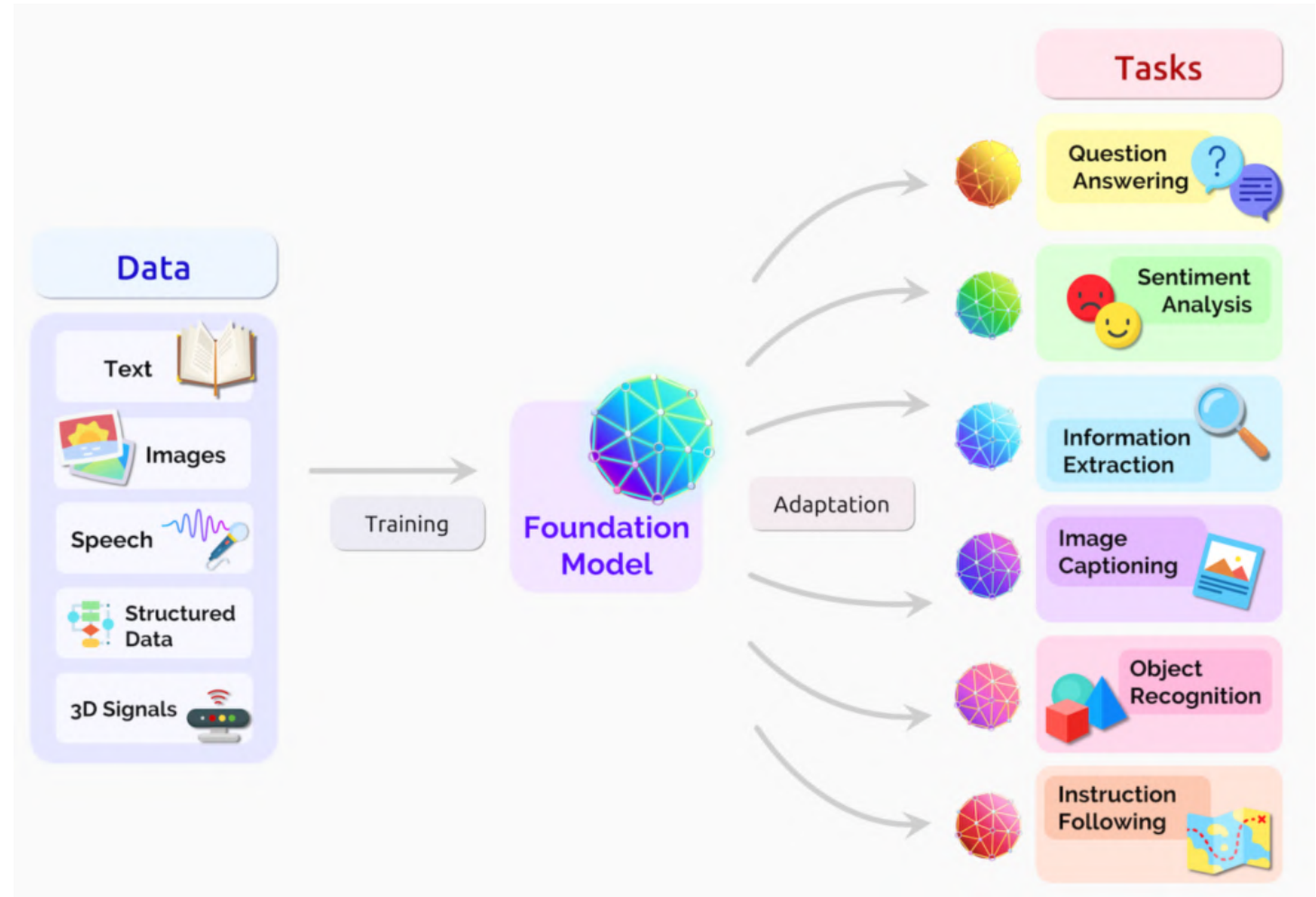
Part I

Introduction to Foundation Models

Foundation models: Definition

A **foundation model** is any model that is trained on broad data (generally using self-supervision at scale), centralizing the information from various modalities, that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks; current examples include GPT-5, Gemini and DinoV3.

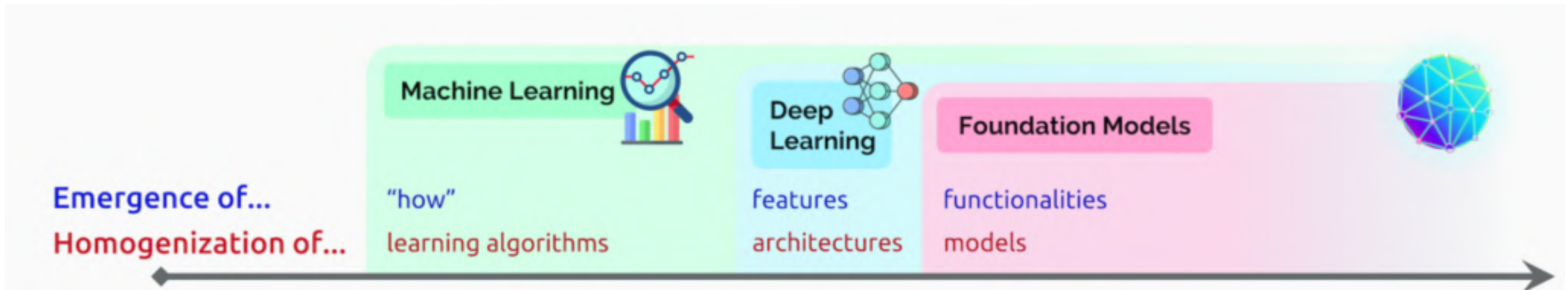
Bommasani, Rishi, et al. "On the opportunities and risks of foundation models." arXiv preprint arXiv:2108.07258 (2021).



Foundation models: Significance

The significance of foundation models can be summarized by two concepts:

- **Emergence of functionalities:** the behavior of a system is implicitly induced rather than explicitly constructed
- **Homogenization of models:** consolidation of methodologies for building machine learning systems across a wide range of applications



Bommasani, Rishi, et al. "On the opportunities and risks of foundation models." arXiv:2108.07258 (2021).

Foundation models: Scale

Transfer learning is what makes foundation models possible, but scale is what makes them powerful. Scale required three ingredients:

- Improvements in **computer hardware** — e.g., GPU throughput and memory have increased $10\times$ over the last four years (*also, the stock of computing power from NVIDIA chips is doubling every 10 months*).
- The development of the **Transformer** model architecture to train much more expressive models
- The availability of much **more training data**. E.g. GPT-3, with 175 billion parameters compared to GPT-2's 1.5 billion, permits in-context learning

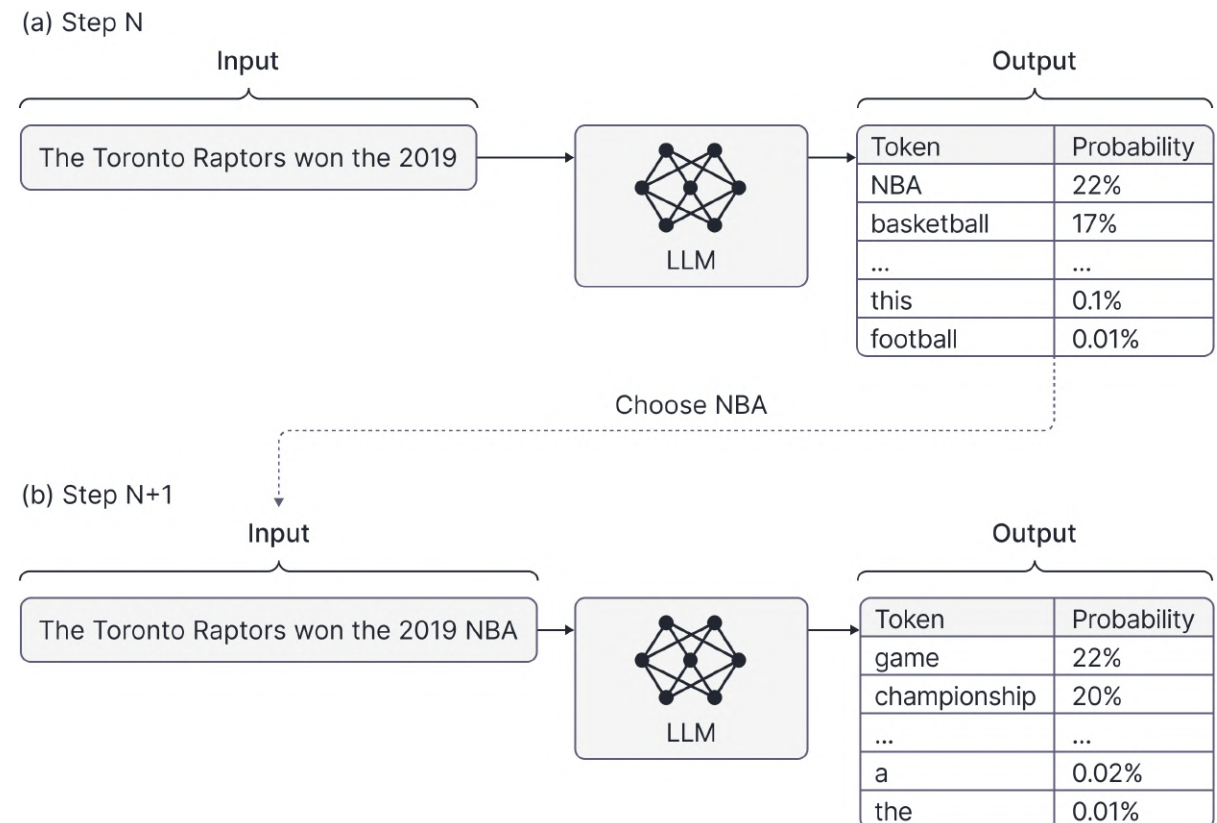
Self-supervised tasks are not only more scalable, only depending on unlabeled data, but they are potentially richer and more useful than models trained on a more limited label space

Bommasani, Rishi, et al. "On the opportunities and risks of foundation models." arXiv:2108.07258 (2021)

Large Language Models: Introduction

- They are systems trained to process input text (*prompt*) and generate output text (response)
- Trained with enormous amounts of unlabeled text, which allows them to acquire logical capabilities and broad knowledge of the world
- Scale: Number of parameters: $> \approx 10$ billion
- Most LLMs (like the GPT family) are **auto-regressive**: they predict the next word based on the previous text

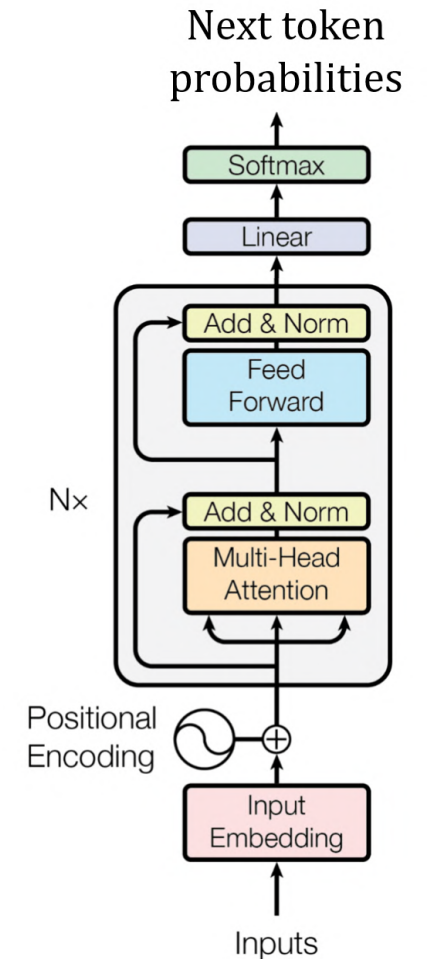
RBC Borealis Research Blog: <https://rbcborealis.com/research-blogs/a-high-level-overview-of-large-language-models/>



GPT-3

Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art finetuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting

Brown, Tom, et al. "Language models are few-shot learners." Advances in neural information processing systems 33 (2020): 1877-1901.

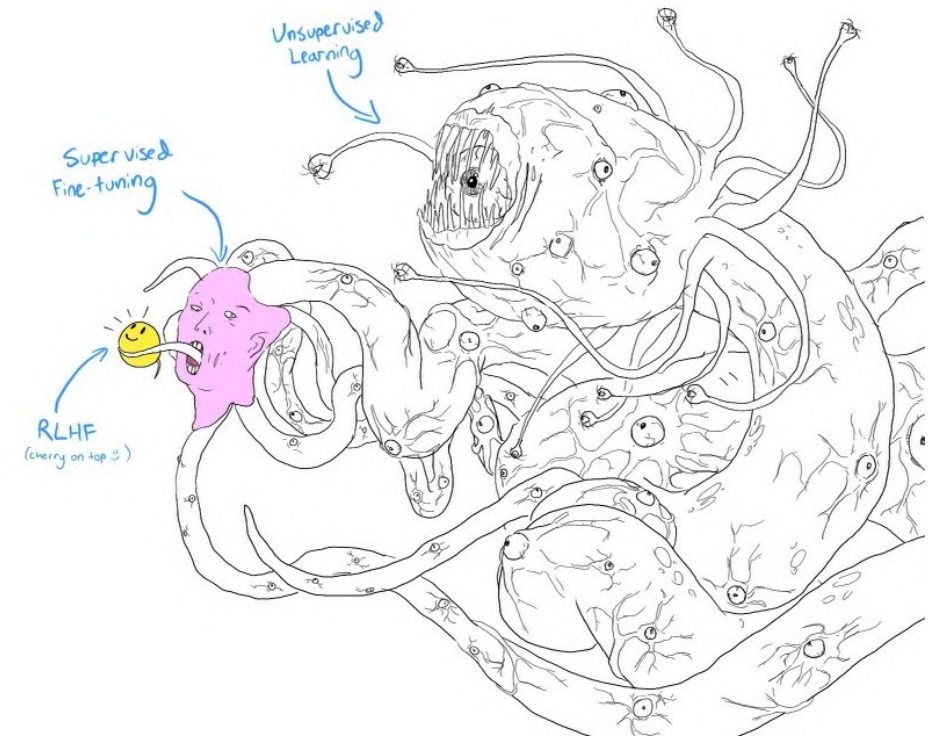


ChatGPT series

The original ChatGPT was a conversational model based on the refinement of the **GPT3** series with advanced *fine-tuning* techniques and **RLHF** (Reinforcement Learning from Human Feedback) as shown in **InstructGPT**

Brown, Tom, et al. "Language models are few-shot learners." Advances in neural information processing systems 33 (2020): 1877-1901.

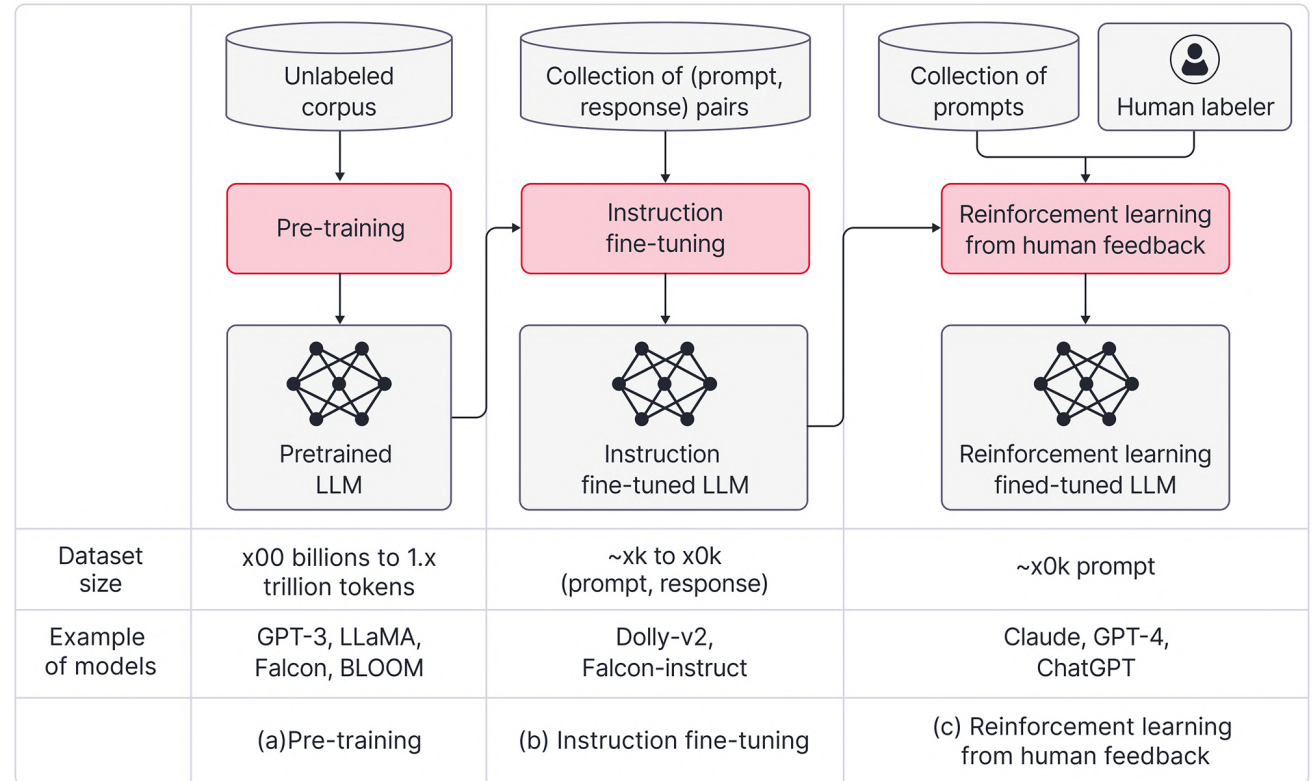
Ouyang, Long, et al. "Training language models to follow instructions with human feedback." Advances in neural information processing systems 35 (2022): 27730-27744.



ChatGPT according to Helen Toner on X,
<https://twitter.com/hlntnr/status/1632030600944033792>

Large Language Models: Three-step training process

- **Pretraining (mandatory):** self-supervised learning of the model with large amounts of text
- **Instruction fine-tuning (optional):** improves the ability to follow specific instructions
- **Reinforcement Learning from Human Feedback - RLHF (optional):** refines behavior according to human preferences



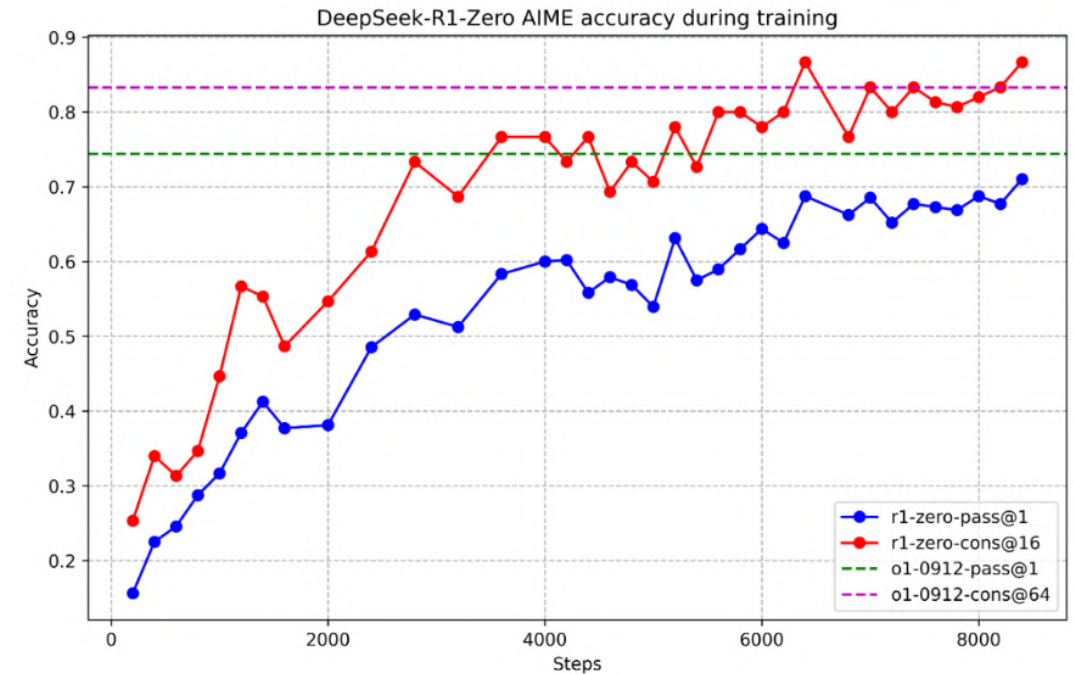
Thinking, the DeepSeek R1 revolution

DeepSeek R1 revolutionized the field: model reasoning through pure reinforcement learning

Key Innovation: No relying on distillation from proprietary models (e.g., GPT-4) or expensive human-labeled Chain of Thought (CoT) data. DeepSeek R1 discovers reasoning patterns emergently through RL, making advanced reasoning capabilities more accessible and reproducible.

Pure RL becomes a crucial step in following research

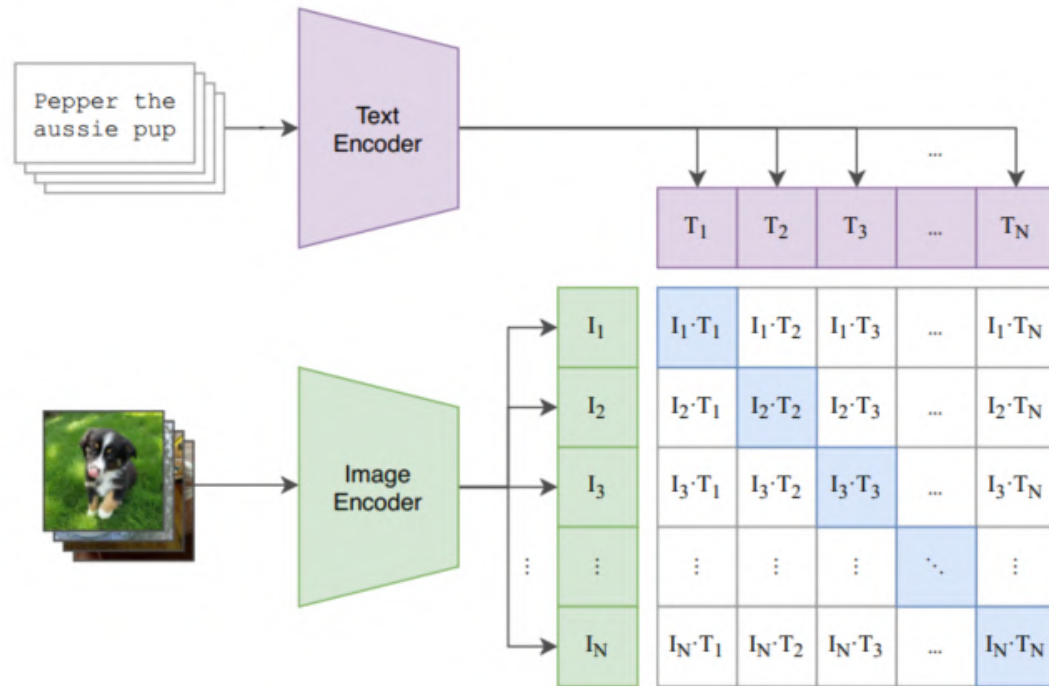
Guo, Daya, et al. "Deepseek-r1 incentivizes reasoning in llms through reinforcement learning." Nature 645.8081 (2025):633-638.



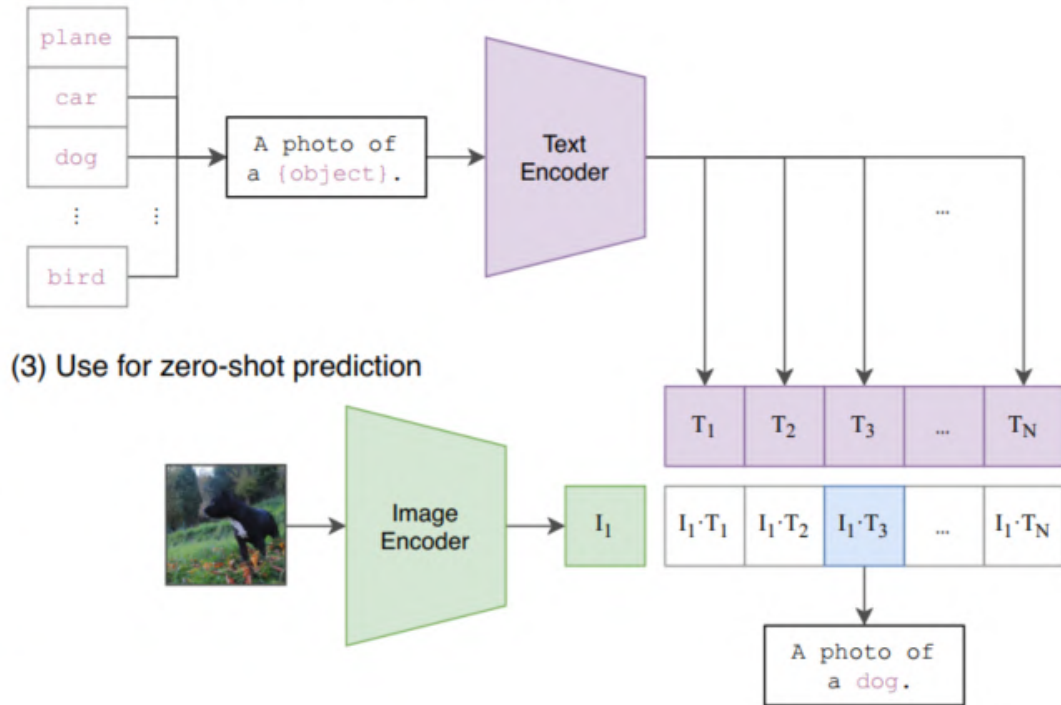
Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444

CLIP: Contrastive Language - Image Pretrain

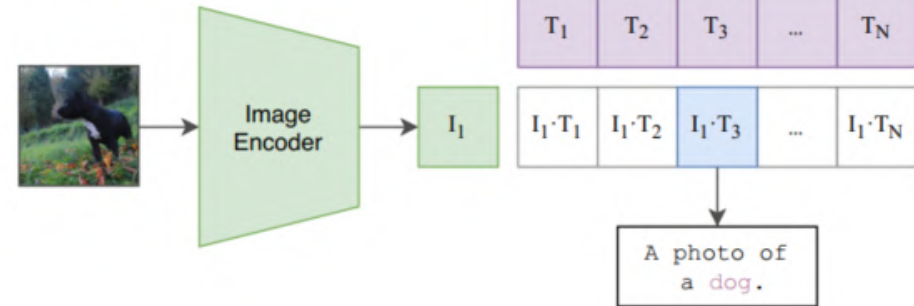
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.

CLIP

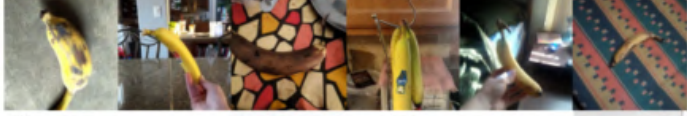
```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

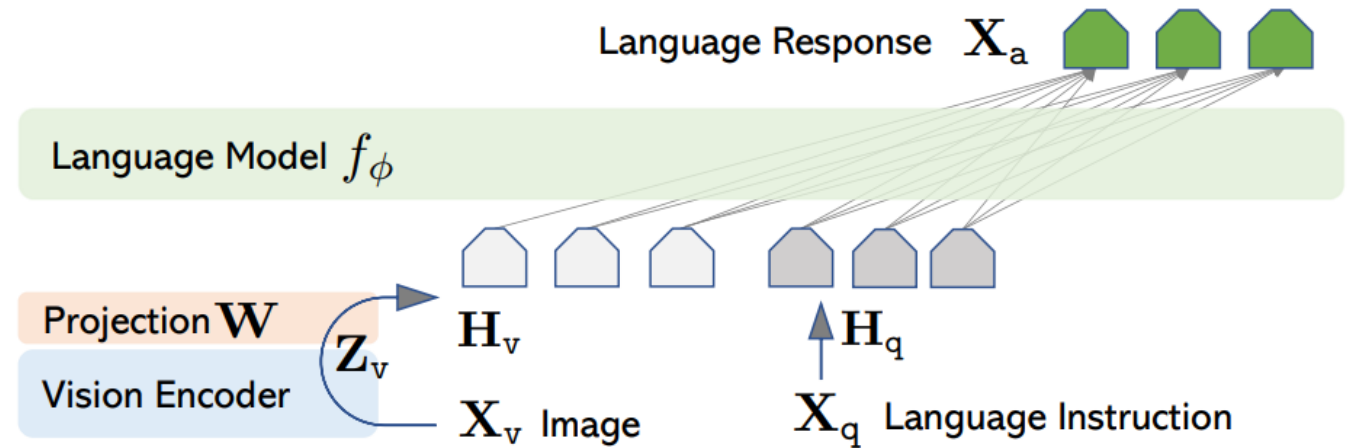
	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%

Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.

LLMs: LLaVA

LLaVA (Large Language-and-Vision Assistant) pioneered the simple yet effective recipe for visual instruction tuning:

1. **Vision Encoder:** Uses a pre-trained CLIP (ViT-L/14) to encode images.
2. **Projection:** A simple linear layer maps vision features to the LLM's embedding space.
3. **LLM Backbone:** A standard LLM (e.g., Vicuna, LLaMA) processes the visual tokens as if they were text.



Liu, Haotian, et al. "Visual instruction tuning." NeurIPS (2023).

Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is.



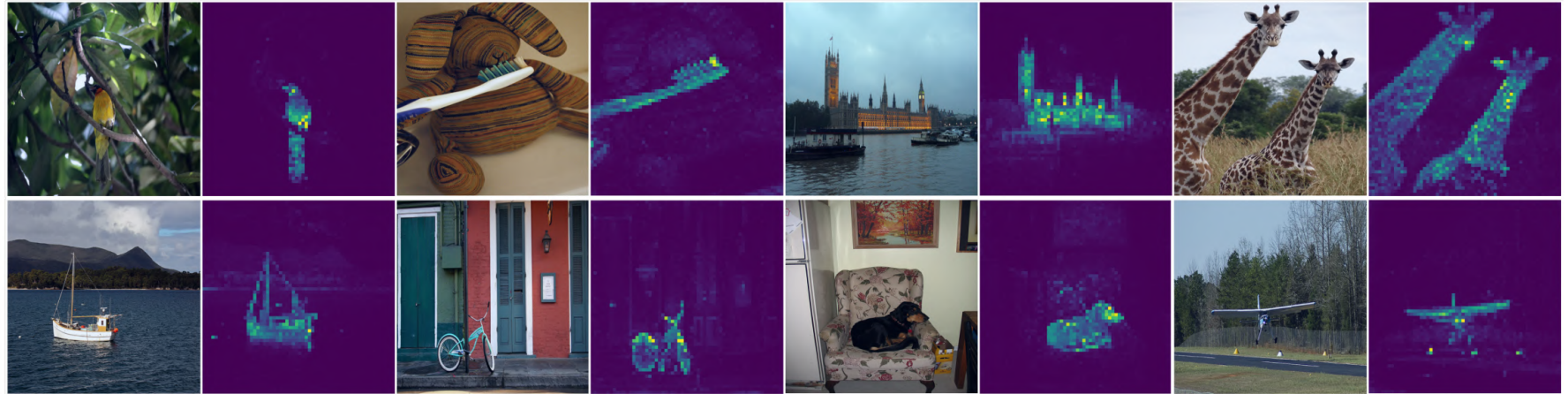
Can you explain this meme in detail?

The meme in the image is a creative and humorous take on food, with a focus on chicken nuggets, as the center of the universe. The meme begins with the phrase "Sometimes I just look at pictures of the Earth from space and I marvel at how beautiful it all is..." The main image features a tray of cooked chicken nuggets, with the chicken pieces positioned to resemble the continents and islands on a world map. The meme playfully suggests that the chicken nuggets represent the Earth, and the various locations depicted in the photo are actually chicken nugget versions of different places. The punchline of the meme appears in smaller text at the bottom, reading "I mean, it's not the real Earth, but how beautiful it is all is." This lighthearted and entertaining meme combines food and humor, using the image of chicken nuggets to create a funny and imaginative representation of the world.

DINOv1: Self-distillation with no labels

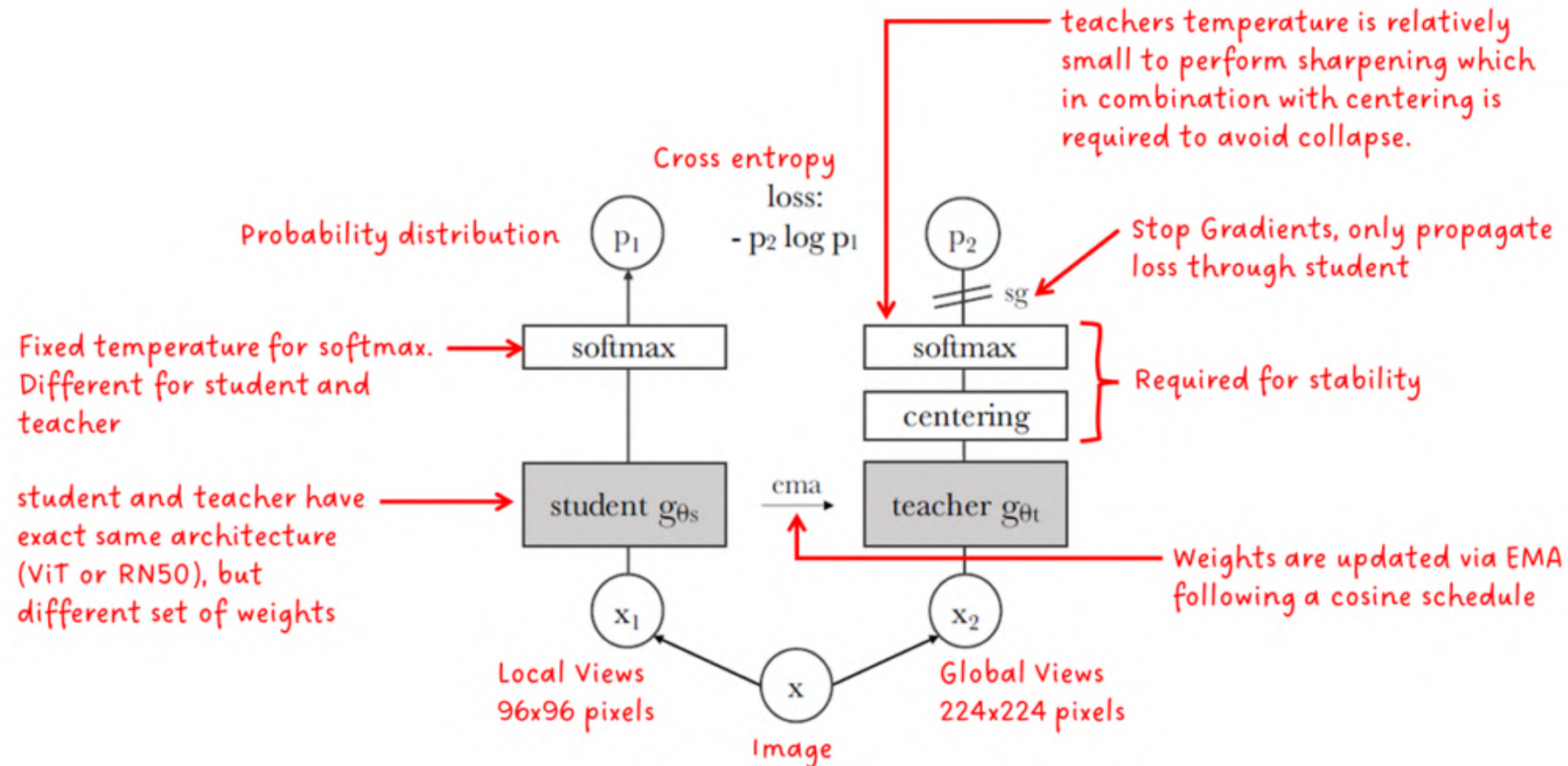
DINO v1:

Demonstrates that ViTs can learn rich visual features without labels through self-supervision using a student-teacher framework



Method: Student network learns to match teacher's output on different augmented views of the same image, where the teacher is an exponential moving average of the student, creating a momentum encoder

DINOv1: $\mathcal{L}_{DINO}$



$\mathcal{L}_{DINO}$: Cross-entropy loss between student & teacher predictions on different crops. Uses centering and sharpening to stabilize training and prevent trivial solutions

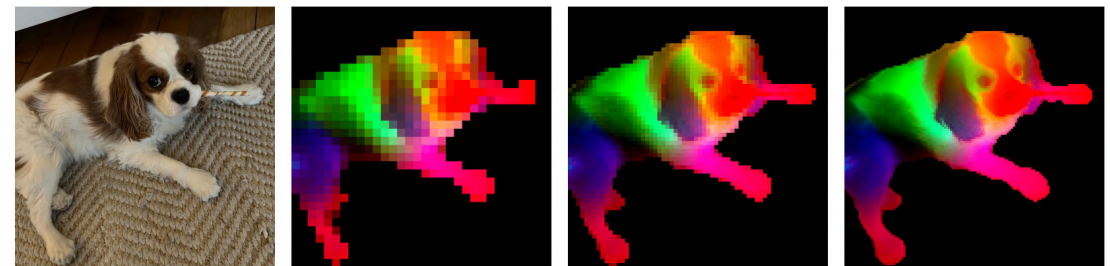
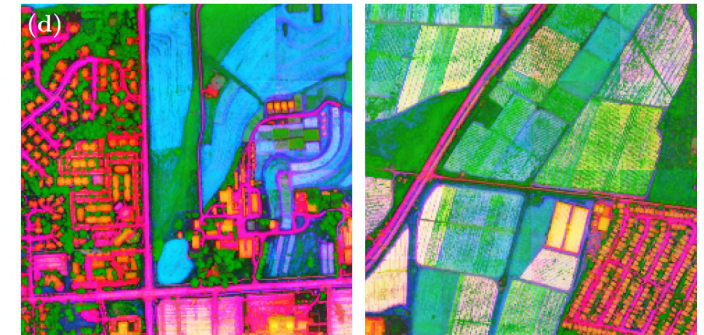
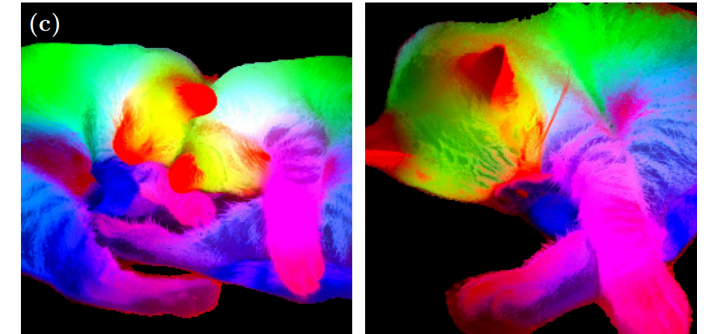
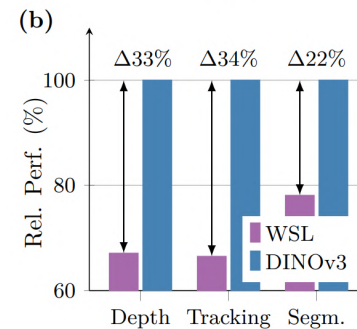
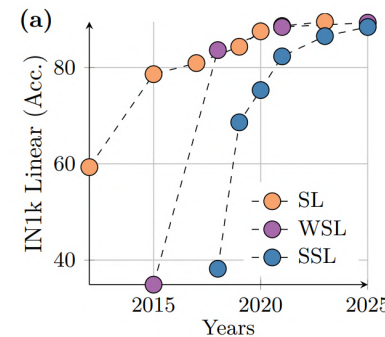
DINOv3: Key Improvements over DINOv1/v2

1. Gram Anchoring for Dense Features: Solves the degradation of dense feature maps that occurs when scaling to large models, enabling high-quality similarity maps at high resolutions (e.g., 4096×4096)

2. Massive Scale: 7B parameter frontier model, distilled into practical ViT-S/B/L variants.

3. Versatility: Generalizing across domains (natural images, satellite imagery, medical data)

Siméoni, Oriane, et al. "Dinov3." arXiv:2508.10104 (2025).



DINOv3: Learning Objective

DINOv3 extends DINOv1 with additional losses for both global semantics and fine-grained local details allowing for big scale model:

- $\mathcal{L}_{DINO}$ (**image-level**): Student matches teacher's global representation across augmented views (inherited from DINOv1)
- $\mathcal{L}_{LiBOT}$ (**patch-level**): Reconstructs masked patch features for local spatial understanding
- $\mathcal{L}_{Gram}$: Enforces patch-level similarity structure by anchoring to an early teacher's Gram matrix, preventing dense feature degradation during scaling
- $\mathcal{L}_{DKoleo}$ (**regularizer**): Spreads features uniformly in embedding space to maximize information



Part III

Foundation Models in Earth Observation

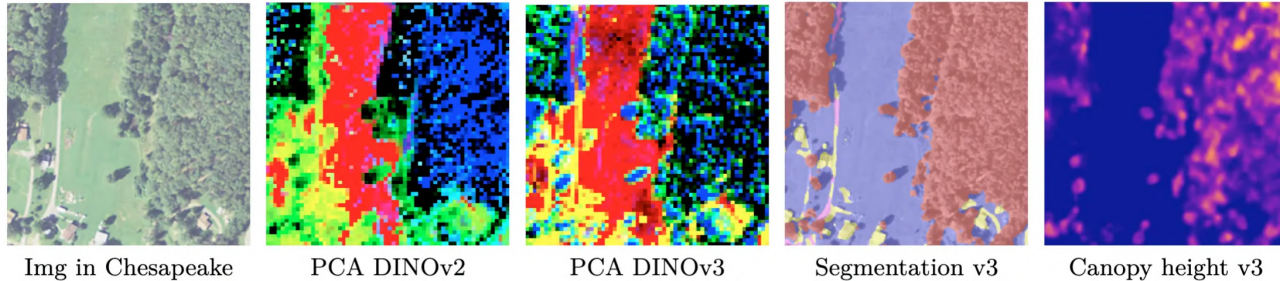
DINOv3 Sat

Table 18: Comparison of our DINOv3 models against strong baselines DOFA (Xiong et al., 2024), Prithvi v2 (Szwarcman et al., 2024), and Tolan et al. (2024) in Geo-Bench tasks. While Privthi-v2 and DOFA leverag all available optical bands, our models achieve significantly better performance with only RGB inputs.

(a) Classification tasks.										
Method	Arch.	FT	Bands	m-BEnet	m-brick-kiln	m-eurosat	m-forestnet	m-pv4ger	m-so2sat	Mean
DOFA	ViT-L	👉	all	68.7	98.4	96.6	55.7	98.2	61.6	79.9
Best of Prithvi-v2	ViT-L/H	👉	all	71.2	98.8	96.4	54.1	98.1	59.1	79.6
Tolan et al. (2024)	ViT-L	🌟	RGB	66.0	97.1	95.2	56.3	94.3	58.1	77.8
DINOv3 Sat	ViT-L	🌟	RGB	73.0	96.5	94.1	60.6	96.0	57.4	79.6
DINOv3 Sat	7B	🌟	RGB	74.0	97.2	94.8	62.3	96.1	62.1	81.1
DINOv3 Web	7B	🌟	RGB	74.6	97.7	97.0	57.9	98.3	63.8	81.6

(b) Segmentation tasks.										
Method	Arch.	FT	Bands	m-cashew*	m-chesapeake	m-NeonTree	m-nz-cattle	m-pv4ger-seg	m-SA-crop	Mean
DOFA	ViT-L	👉	all	81.2	61.6	58.5	77.4	95.1	35.7	68.3
Best of Prithvi-v2	ViT-L/H	👉	all	90.2	69.4	59.1	81.0	95.3	41.9	72.8
Tolan et al. (2024)	ViT-L	🌟	RGB	92.8	73.7	58.1	83.1	94.7	35.1	72.9
DINOv3 Sat	ViT-L	🌟	RGB	94.2	75.6	61.8	83.7	95.2	36.8	74.5
DINOv3 Sat	7B	🌟	RGB	94.1	76.6	62.6	83.4	95.5	37.6	75.0
DINOv3 Web	7B	🌟	RGB	96.0	76.5	66.4	83.7	95.9	36.8	75.9

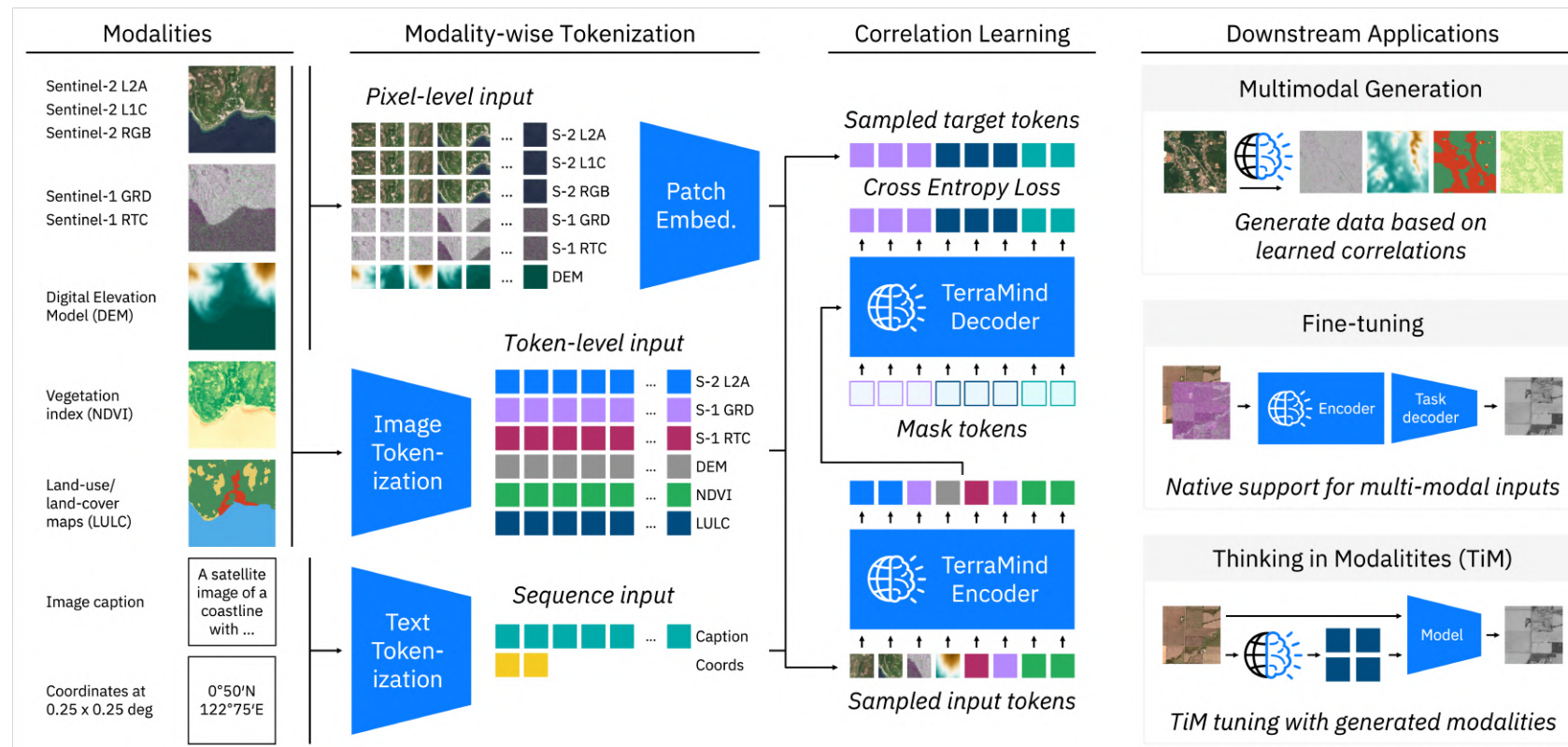
* Conversion to 6 classes following Szwarcman et al. (2024).



- **Pretrained on SAT-493M**, a dataset of 493 millions of 512×512 images sampled randomly from Maxar RGB ortho-rectified imagery at 0.6 meter resolution.
- **Versatility of both SAT and WEB model:** DINOv3 SAT excels in metric tasks like depth estimation, leveraging satellite-specific priors. In contrast, the DINOv3 web model achieves state-of-the-art results on semantic geospatial tasks through diverse, universal representations.

Terramind

First any-to-any generative, and large-scale multimodal model for Earth observation pre-trained on 500 billion tokens from global geospatial data The model digests multi-scale representations at pixel-level and token-level simultaneously.



Blumenstiel, B., Jakubik, J., Marsocci, V., et al. (2025). TerraMind: Large-Scale Generative Multimodality for Earth

Observation. arXiv preprint arXiv:2504.11171.

Terramind

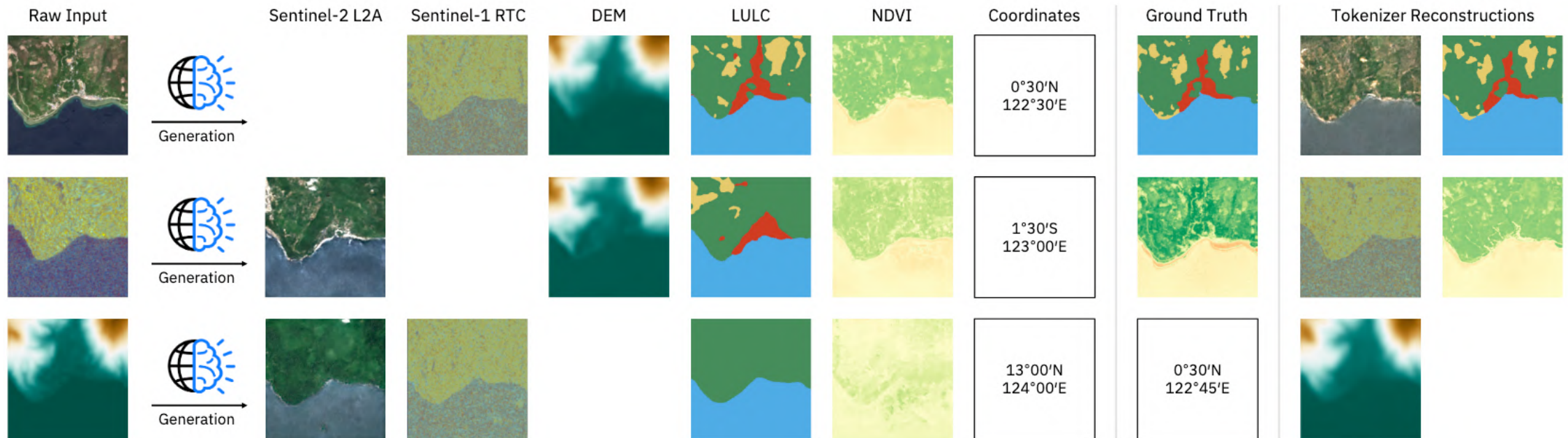


Figure 3. Chained generation example of TerraMindv1-B starting from either optical, radar, or digital elevation data. Left is input, middle is artificially generated data by TerraMind, right represents ground truths and tokenizer reconstructions, respectively.

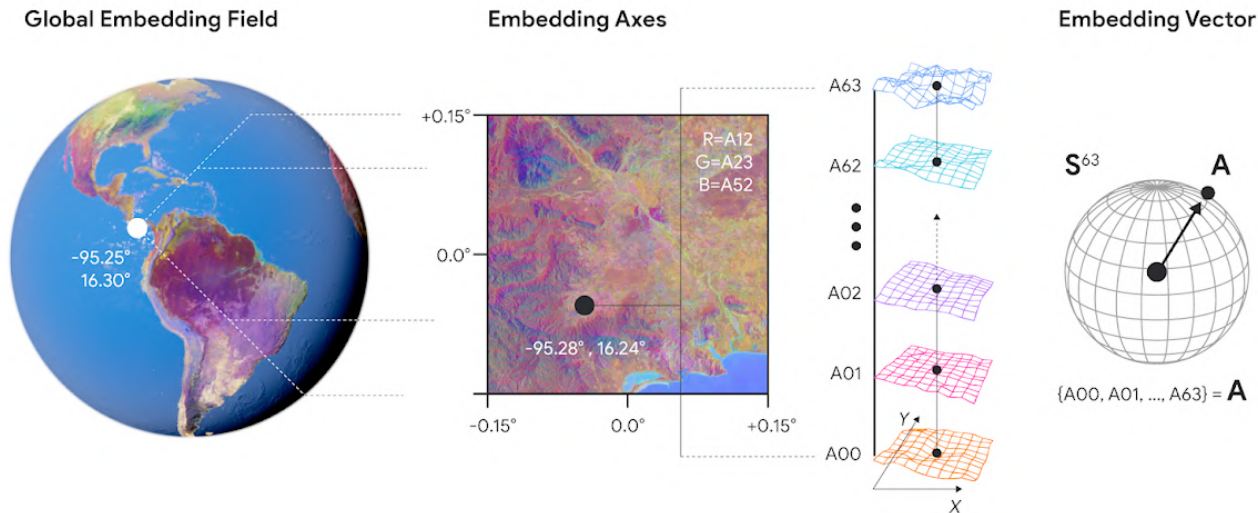
Terramind: Thinking in Modalities (TiM)

- Thinking in Modalities (TiM) is a **recursive fine-tuning and inference technique** designed to enhance multimodal learning by leveraging the generative capabilities of the model itself.
- Chain of thought for modalities:** This formulation allows the model to reason over both observed and inferred modalities effectively enriching the input space

Fine-Tuning	Input	$\text{IoU}_{\text{Water}}$	mIoU
TerraMindv1-B	S-1	68.00	81.06
TerraMindv1-B	S-2	82.26	89.70
TerraMindv1-B TiM	S-1 + <i>gen. LULC</i>	72.25	83.65
TerraMindv1-B TiM	S-2 + <i>gen. LULC</i>	84.75	91.14

Table 8. Thinking-in-modalities (TiM) tuning compared with standard full fine-tuning approaches on the Sen1Floods11 dataset.

Alpha Earth

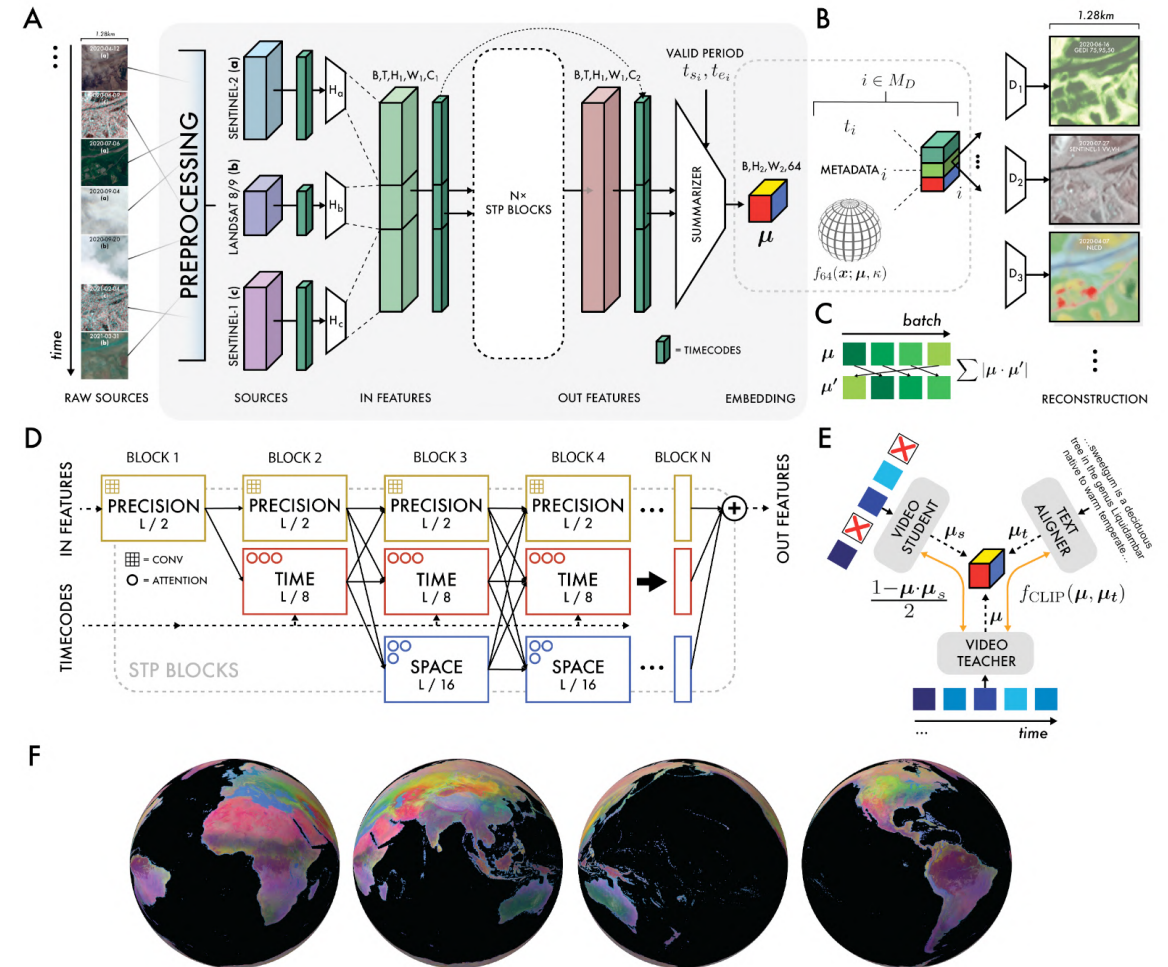


- A multi-modal "**virtual satellite**" via representation learning.
- Fuses heterogeneous data into a unified, light, analysis-ready embedding.
- Includes all terrestrial positions (sphere) with an annual **temporal component** from various different earth observation **modalities** (>3 billion observation frames): optical, radar, lidar, gravity fields with a **resolution of 10x10 meter squares**.

Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., et al. (2025). AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*.

AlphaEarth

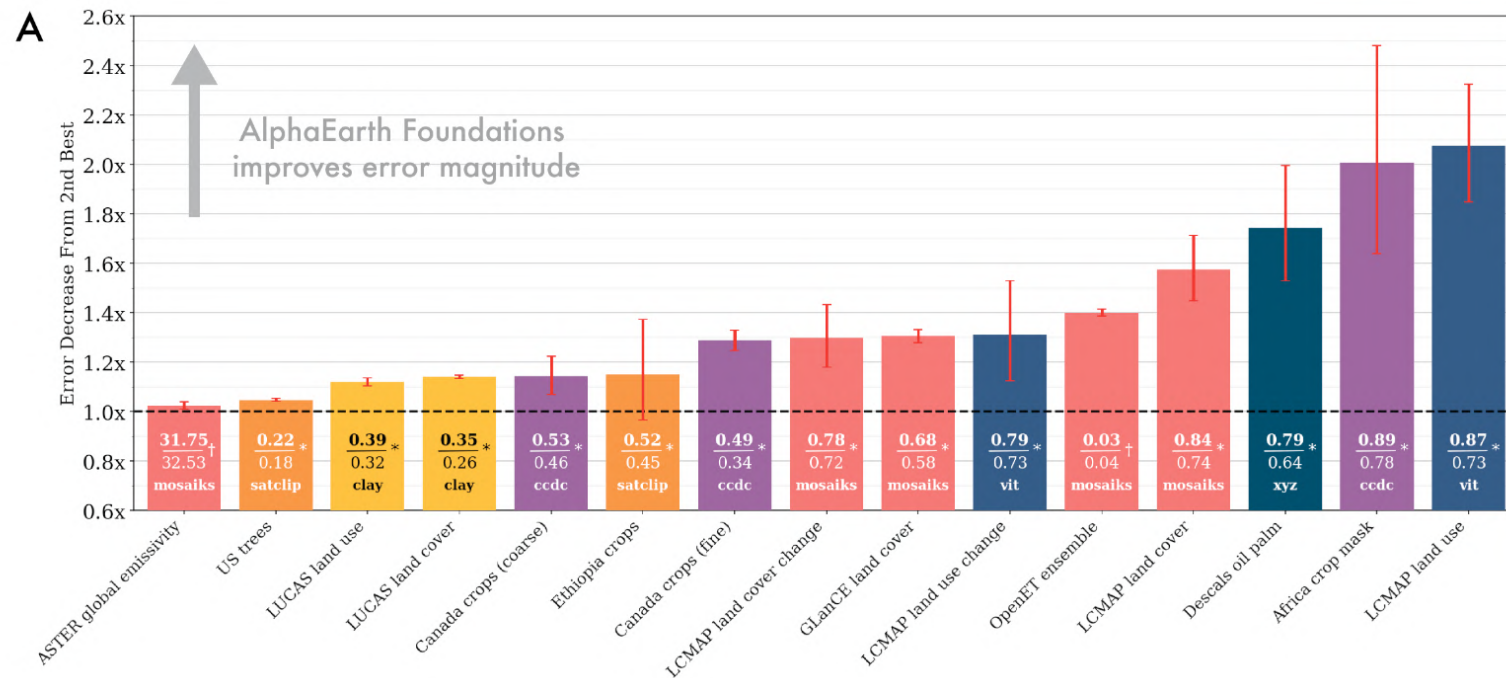
- **STP attention blocks:** Combines spatial self-attention (ViT-style) to capture local texture/geometry with time-axial attention to model temporal dynamics and periodicity (phenology) and uses CNN's to align different resolutions.
- **Teacher-Student and text encoder Consistency:** Contrastive learning between the video teacher and student model, and text encoder. This enforces robustness against missing modalities and sensor noise.



AlphaEarth

AlphaEarth achieves a 23.9% reduction in error rates on average across 15 diverse downstream tasks

- Demonstrates state of the art capabilities for one/few-shot learning.
- By fusing SAR and optical data in latent space, reconstructs successfully areas with permanent cloud cover.
- Integrating AlphaEarth embeddings with socio-economic data (Population Dynamics) improved R2 performance on FEMA risk indices by 11%.



OlmoEarth

- **Stable Latent Image Modeling (SLIM)** We replace the target encoder of Latent MIM with a linear projection from image patches to token space that is randomly initialized and never updated during training. This simple modification stabilizes training but maintains the representative power of modeling in latent space.
- **Modality-aware masking:** random token masking with full modality reconstruction. Earth images are too redundant.
- Optical + Radar + Semantic Maps
- When evaluating embeddings OlmoEarth achieves the best performance on 15 out of 24 tasks, and with full fine-tuning it is the best on 19 of 29 tasks
- **Open** end to end and provides "low code" fine-tuning interface

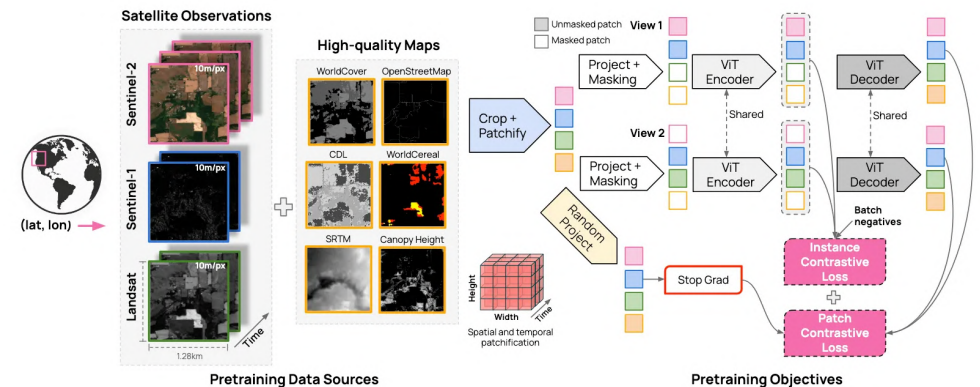


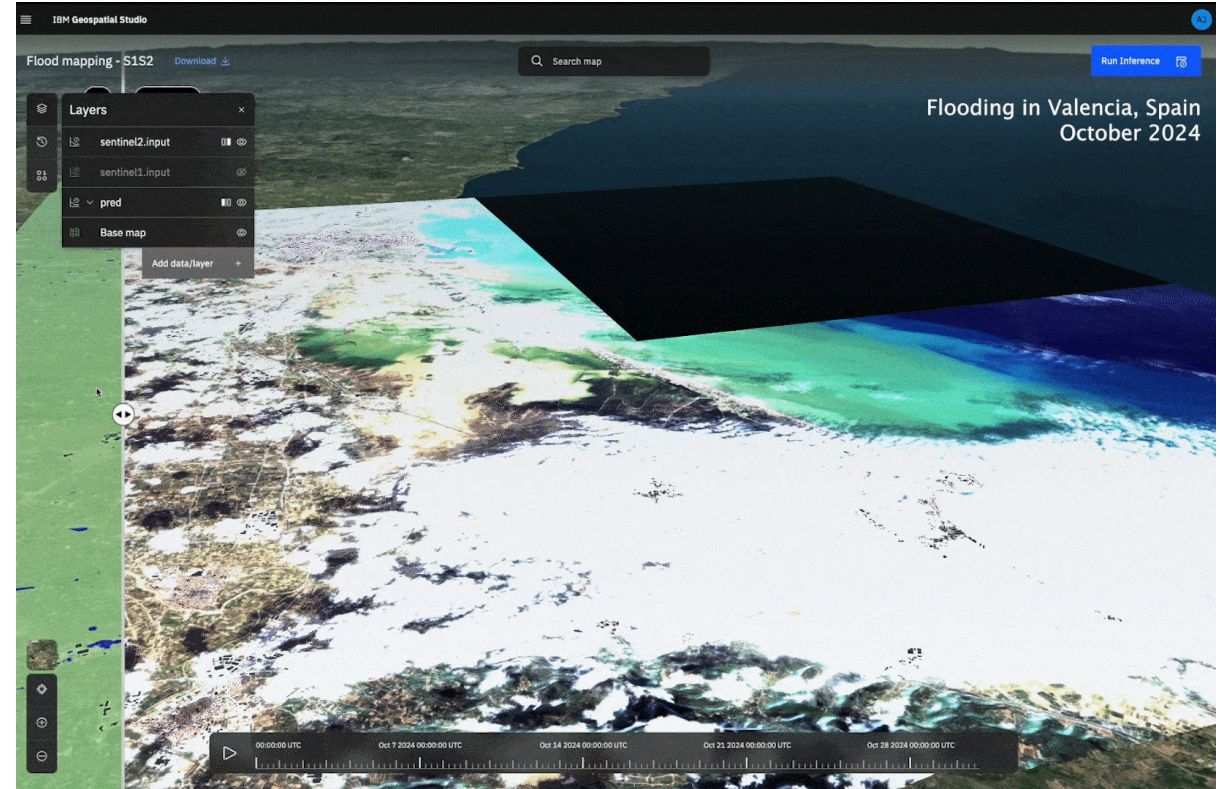
Figure 3 We train OlmoEarth with a combination of satellite observations and high-quality maps. After tokenizing these inputs, we: (1) apply a modality-aware masking strategy to define which tokens are inputs vs. targets, (2) pass the target tokens through fixed random projections to construct targets, (3) pass the input tokens through our learned encoders, and then (4) through a decoder which predicts the target tokens and (5) apply a modality-aware patch discrimination loss between the predicted and target tokens. Steps 1-5 are applied twice on the same data to then (6) apply an instance contrastive loss over the aggregated tokens per instance.

Herzog, H., Bastani, F., Zhang, Y., Tseng, G., Redmon, J., et al. (2025). OlmoEarth: Stable Latent Image Modeling for Multimodal Earth Observation. arXiv preprint arXiv:2511.13655.

Prithvi 2.0

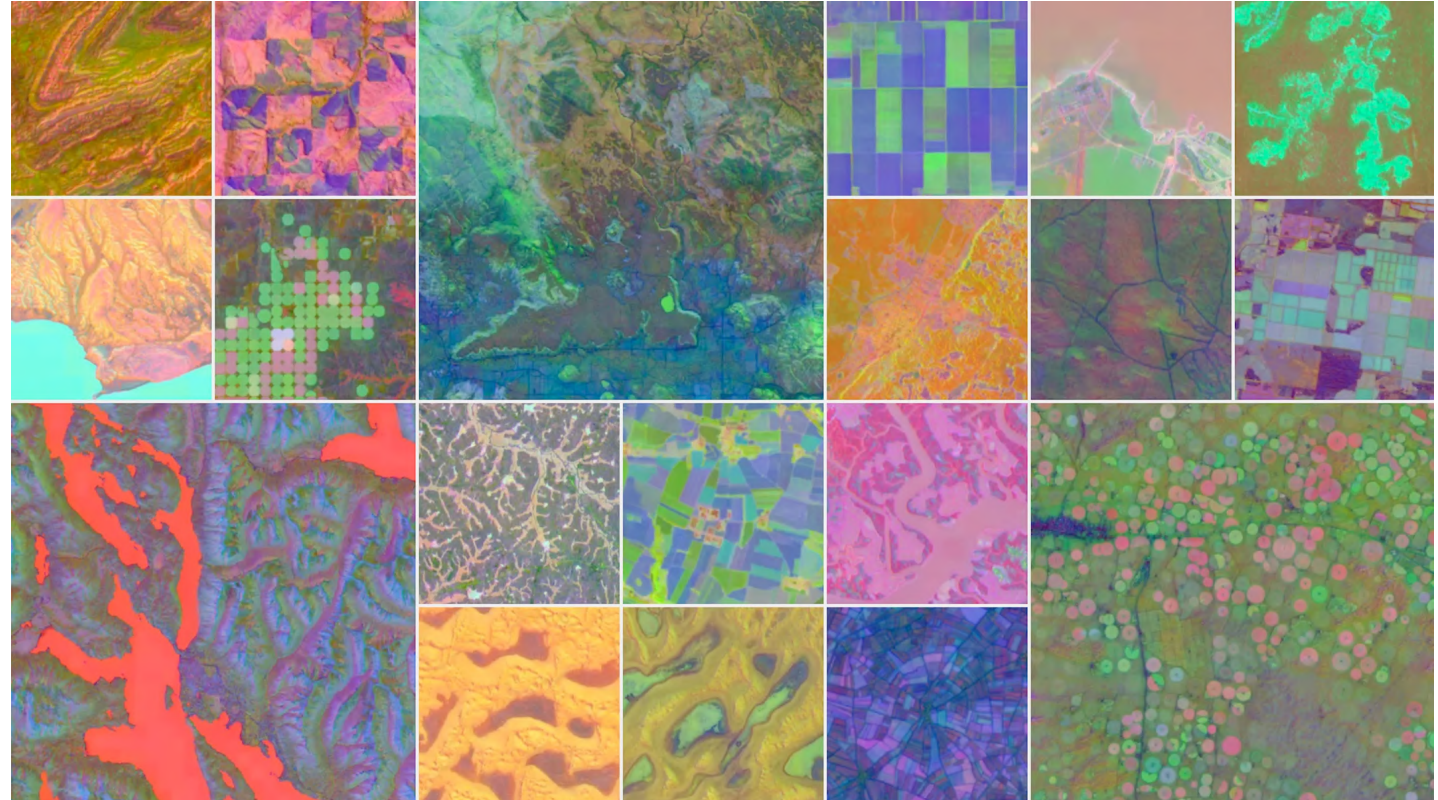
- **3D Tubelet Embeddings** processing time, height, and width simultaneously.
- Encodes Geolocation (lat/lon) and Date (day-of-year) using **Sinusoidal Encodings** and injects them directly into the encoder.
- **600M Parameter Scale**
- A "Tiny" version (**5M params**) exists that runs on-orbit, showing only a <4% performance drop compared to the 100M version.

Szwarcman, D., Roy, S., Fraccaro, P., Gíslason, P. E., Blumenstiel, B., Ghosal, R., et al. (2024). Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation Applications. *arXiv preprint arXiv:2412.02732*.



Other state of the art EO FM models

- **Panopticon**: Advancing Any-Sensor Foundation Models for Earth Observation
- **CROMA**: Remote Sensing Representations with Contrastive Radar-Optical Masked Autoencoders
- **SatCLIP**: Global, General-Purpose Location Embeddings with Satellite Imagery



Conclusions

- **Scale & SSL Dominance:** When data and compute are abundant, **Self-Supervised Learning (SSL)** is proving to be the superior paradigm (e.g., DINOv3 closing the gap with supervised methods).
- **The "Data Scarcity" Question:** In specialized scientific domains, where data might be scarce or fundamentally different, it remains an open question whether general FMs or domain-specific supervised learning will prevail.
- **Multimodal Architecture:** LLaVA (Vision Encoder + Projection + LLM) has established itself as a powerful and flexible paradigm for building highly multimodal FMs.
- **Generative Reasoning:** TerraMind shows us that generating intermediate data (TiM) helps downstream tasks.